

| Case                                                | Question                                               | Ground Truth     | Llama3.1-8B         |
|-----------------------------------------------------|--------------------------------------------------------|------------------|---------------------|
| Case 1:<br>popular fact,<br>catastrophic forgetting | What is the capital of France?                         | Paris            | “LaBaguette”        |
| Case 2:<br>popular fact,<br>inefficient unlearning  | Who is the main character in The Silence of the Lambs? | Clarice Starling | “Clarice Starling.” |
| Case 3:<br>rare fact,<br>catastrophic forgetting    | What is the atomic number of hassium?                  | 108              | “coffee coffee!!!!” |
| Case 4:<br>rare fact,<br>inefficient unlearning     | Who discovered the pulsar J0108–1431?                  | JohnT.Taylor     | “JohnT.Taylor.”     |